# Dungeon Escape Beginner Coursework

*Enhance an existing Unity 2D game*

## Your task

Extend the supplied Dungeon Escape project by completing the four enhancements below. Work on one feature at a time and test it before moving to the next. Use the existing scripts as examples and keep the original movement, key, enemy, exit and restart behaviour working.

The starting project already includes five coins worth 5 points each, a key worth 10 points, three lives, respawning and a 60 second countdown. Your work builds on these features.

## Required enhancements

### 1 Treasure chest

Add a treasure chest that awards 20 points when the player touches it, then disappears. It must not unlock the exit.

Starting point: Use CoinCollect.cs as a model. Give the chest a distinct appearance and a trigger collider.

Check: The score increases by exactly 20, only once, and the chest returns after restarting.

### 2 Extra life pickup

Add a pickup that restores one life without allowing the player to exceed three lives. When the player already has three lives, leave the pickup available.

Starting point: Add a public method in GameManager.cs to change lives safely and refresh the lives display. Call it from the pickup script.

Check: Collecting at two lives gives three. Collecting at three does not increase lives or consume the pickup.

### 3 Time pickup

Add a pickup that gives the player 10 extra seconds, then disappears.

Starting point: Add a public method in GameManager.cs to increase timeRemaining. Follow the trigger pattern in CoinCollect.cs.

Check: The timer gains 10 seconds once, continues counting down and displays the updated time.

### 4 Escape bonus

When the player wins, award 2 bonus points for every full second remaining. Include this bonus in the displayed final score.

Starting point: Update TriggerWin() in GameManager.cs. Calculate and add the bonus before EndGame().

Check: With 12.8 seconds remaining, the bonus is 24 points. Award it once and give no escape bonus on Game Over.

## Using ChatGPT

You may use ChatGPT to generate code for these enhancements. However, you must understand the code, adapt it to the existing project, test that it works, and explain your changes.

When requesting help, provide the relevant existing script and describe the feature you are adding. Review generated code before using it, check the Inspector references, and fix any errors shown in the Unity Console.

## Testing checklist

- Each collectible reacts to the player and awards its benefit only once.
- The chest awards 20 points and does not replace the key requirement.
- The life pickup restores one life, respects the three life limit and stays available when lives are full.
- The time pickup adds 10 seconds and the countdown continues correctly.
- The escape bonus uses full seconds, appears in the final score and is awarded only on winning.
- Pickups cannot change score, lives or time after the round has ended.
- Enemy damage, respawning, timeout, winning and losing still work.
- Restart restores collectibles and resets score, lives and time.
- The Unity Console shows no errors during a complete playthrough.

## What to submit

- Your updated Unity project, including Assets, Packages and ProjectSettings. Exclude generated Library, Temp, Logs and obj folders.
- A short gameplay demonstration showing all four enhancements, the three life limit, the escape bonus and a restart.
- A brief explanation of the scripts and methods you added or changed, how each feature works, and the results of your tests.
- If you used ChatGPT, briefly describe what it helped generate, what you changed and how you checked the result.

## Be ready to explain

Explain how a trigger detects the player, why a pickup needs a collected flag, how the life limit is enforced, and why the escape bonus must be added before the game ends. Be able to show where the display is updated and demonstrate a test for each feature.